



The OLED is connected via the I²C interface (SDA to A4, SCL to A5), and the libraries handle all low-level communication with the display controller. These libraries make it straightforward to display sensor readings, system status, or messages alongside the RGB LED colour effects. The program's `setup()` function initialises the OLED display and sets pin modes for the RGB LED, while the `loop()` function continuously updates the LED colour and OLED content based on the programmed logic.

Construction and testing

The circuit can be assembled either on a breadboard for prototyping or on a general-purpose PCB for permanent use. For breadboard assembly, the Arduino Uno is placed separately, and jumper wires are used to connect the RGB LED and OLED display module, as shown in

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Fig. 2. For PCB assembly, the same connections are soldered, ensuring correct polarity for the LED and proper resistor placement, while keeping track widths adequate for stable connections.

For testing, the source code is uploaded into the Arduino Uno using a USB cable and a laptop or desktop. The assembly is then completed after removing the USB cable. The circuit is powered by a 9V battery connected to the Arduino's DC jack.

During testing, the RGB LED cycles through different colours every 5 seconds as PWM signals are sent to pins 9, 10, and 11, while the OLED simultaneously displays the corresponding colour name. If any colour or display output is incorrect, wiring continuity should be checked, resistor values verified, and the OLED's I²C address confirmed. Once tested and verified, the system operates as intended, providing synchronised LED colour changes and matching text descriptions on the OLED. **EFY**

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